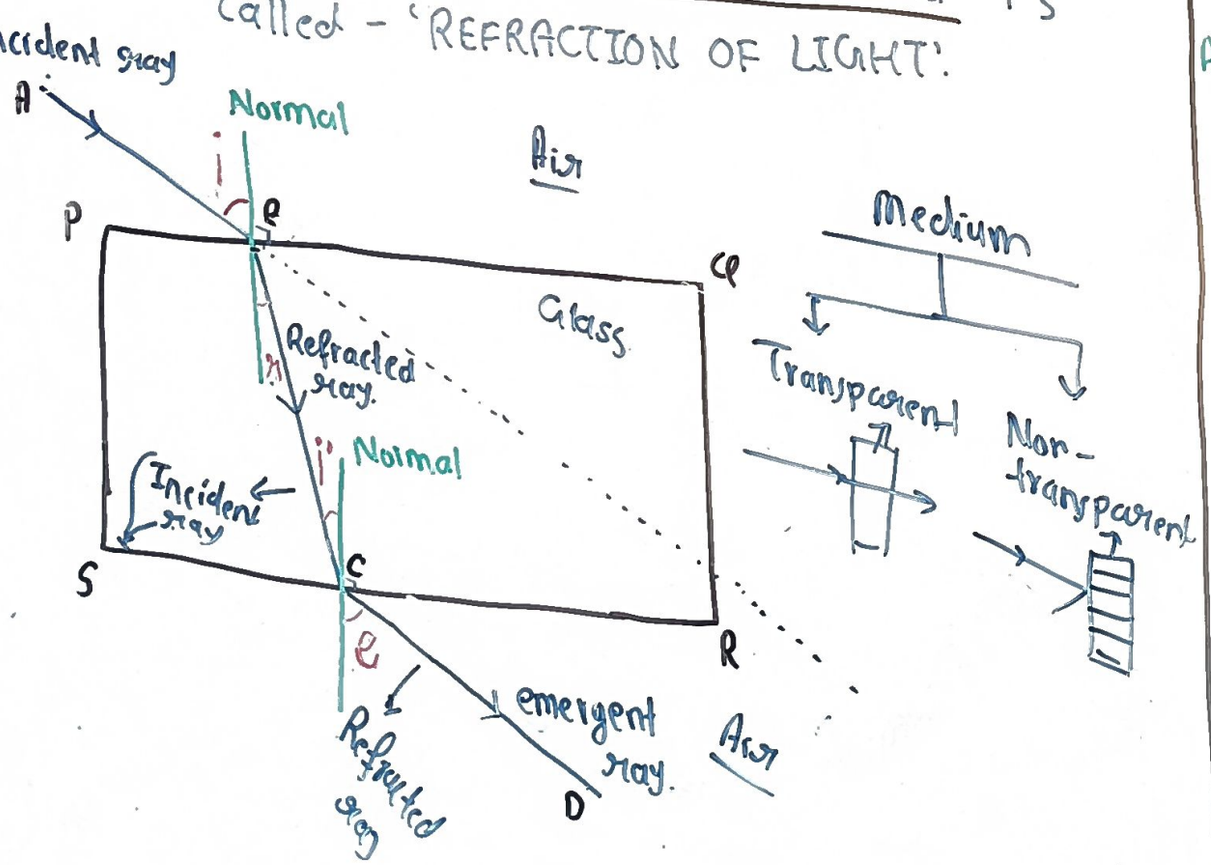
REFRACTION OF LIGHT.

REFRACTION.

1) **Behavior of light :-** Light travels along a straight line within a uniform (same) medium (same density).

2) **Defn :-** ① The change in direction of light ray ② when passing from one transparent medium (e.g. air) to another transparent medium (e.g. glass) at ③ the contact of these two media is called - 'REFRACTION OF LIGHT'.



Here,

Rays
- ① AB → Incident ray for surface PQ
- ② BC → Refracted ray for surface PQ
  → Incident ray for surface RS
- ③ CD → Refracted ray for surface RS
  → Emergent ray for overall setup.

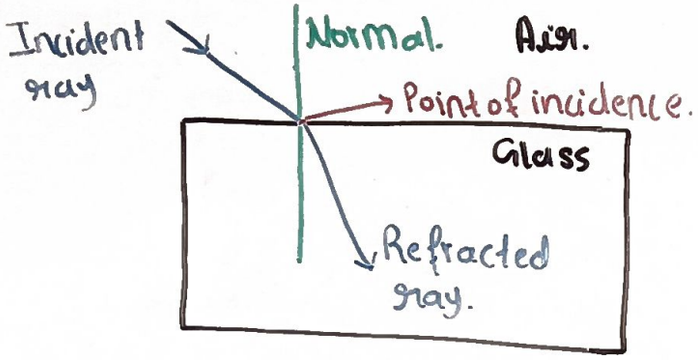
Angles.
- ④ $\angle i$ = Incident angle
- ⑤ $\angle r$ = Refracted angle
- ⑥ $\angle i'$ = Incident angle
- ⑦ $\angle e$ = emergent angle

* $\angle i' = \angle r$ ——— Internal alternate angles

* $\angle e = \angle i$ ——— External alternate angles.

LAWS OF REFRACTION.

LAW-1 : ① Incident ray, refracted ray & Normal lie at the same point i.e. point of incidence



② Incident ray, refracted ray & normal, all lie in the same place.

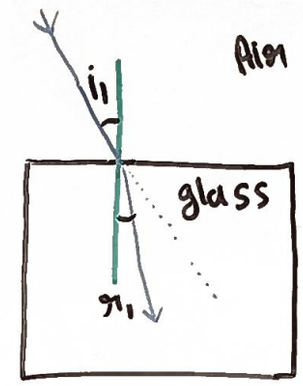
③ The incident ray & refracted are on the opposite side of normal.

LAW-2 : Snell's Law

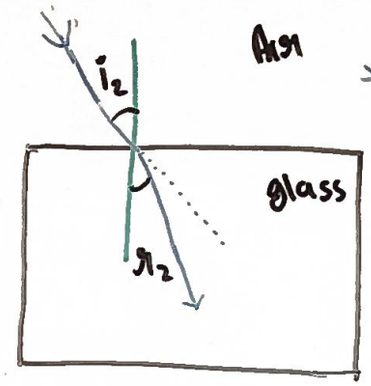
For a given pair of media (e.g. water-glass, water-diamond, kerosene-diamond.), the ratio of the sine of incident angle to the sine of the refracted angle is constant.

$$\frac{\sin i}{\sin r} = \text{constant} = n.$$

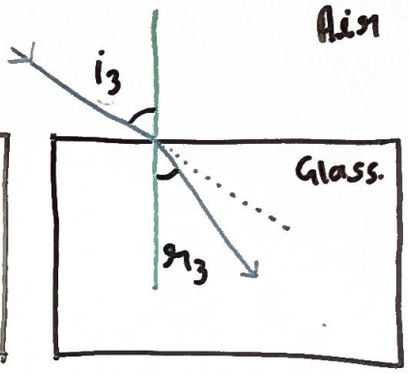
$n \rightarrow$ Refractive index.



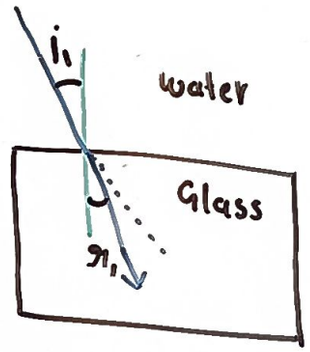
$$\frac{\sin i_1}{\sin r_1} = n = 1.41$$



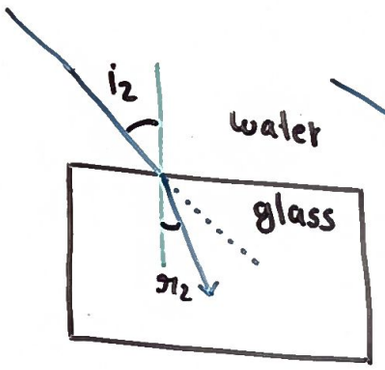
$$\frac{\sin i_2}{\sin r_2} = n = 1.41$$



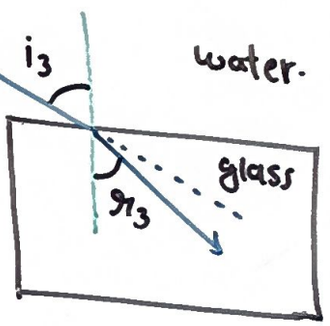
$$\frac{\sin i_3}{\sin r_3} = n = 1.41$$



$$\frac{\sin i_1}{\sin r_2} = n = 1.2$$

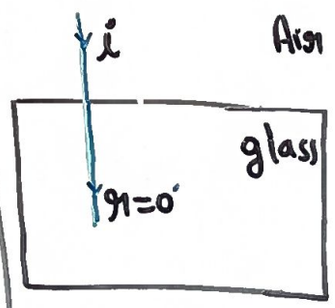


$$\frac{\sin i_2}{\sin r_2} = n = 1.2$$



$$\frac{\sin i_3}{\sin r_3} = n = 1.2.$$

Special case : ⟶



$$\frac{\sin i}{\sin r} = n$$

$$\frac{\sin i}{\sin 0°} = 1.41$$

$$\frac{\sin i}{0} = 1.41$$

$\sin i = 0$

$\therefore i = \sin^{-1}(0)$

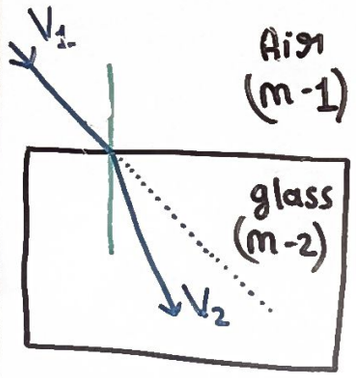
$\boxed{i = 0}$

$\because i = r = 0$

Light do not refract when incident normally to surface

REFRACTIVE INDEX.

① Defⁿ :- The ratio of speed of light in med-1 to speed of light in medium-2 is called 'RI'.



② The velocity of light is different in different media.

$$^1\eta_2 = \frac{V_1}{V_2}$$

- V_1 ⇨ Speed of light in medium-1
- V_2 ⇨ Speed of light in medium-2
- $^1\eta_2$ ⇨ RI of med-2 w.r.t. med-1.

$$^2\eta_1 = \frac{V_2}{V_1}$$

- V_2 ⇨ Speed of light in med-2
- V_1 ⇨ Speed of light in med-1
- $^2\eta_1$ ⇨ RI of med-1 w.r.t. med-2.

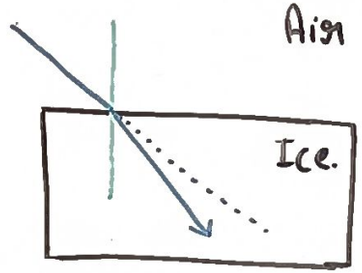
∴

Notes on REFRACTIVE INDEX.

① R.I. tells us extent of bending of light in different media.

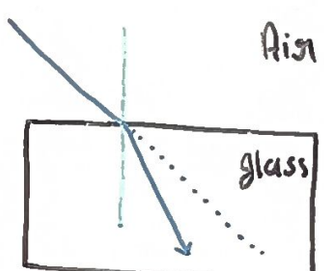
② $\eta \propto$ bending of light.

③ More R.I. ⟹ More bending.



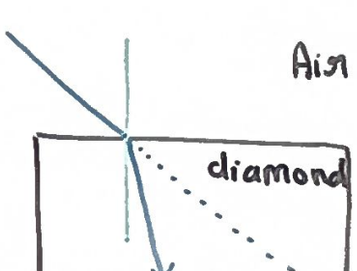
$^a\eta_i = 1.31$

Bending :- ↑



$^a\eta_g = 1.66$

Bending :- ↑

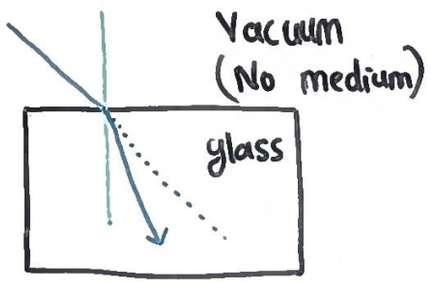


$^a\eta_d = 2.42$

Bending :- ↑

ABSOLUTE REFRACTIVE INDEX.

Defⁿ :- The R.I. of any medium w.r.t. vacuum is called 'absolute refractive index.'



$$\eta = \frac{c}{v}$$

- c ⇨ Velocity of light in vacuum
- v ⇨ Velocity of light in medium
- n ⇨ Absolute R.I

* Rarer medium :- The medium having low density.

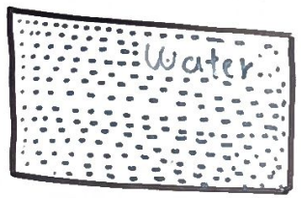
→ ↓ Density ⇨ ↓ R.I. ⇨ ↓ Bending.



घना

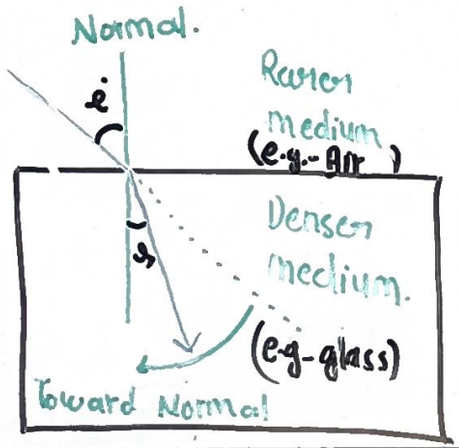
* Denser medium :- The medium having high density.

→ ↑ Density ⇨ ↑ R.I. ⇨ ↑ Bending.



DIRECTION OF BENDING.

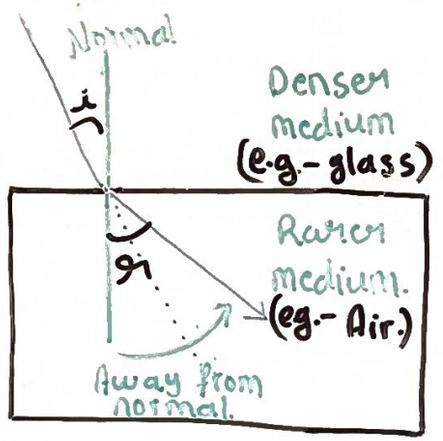
1. Rarer medium → (light) → Denser medium



* When light travels from rarer medium to denser med. it bends toward 'Normal.'

* $r < i$

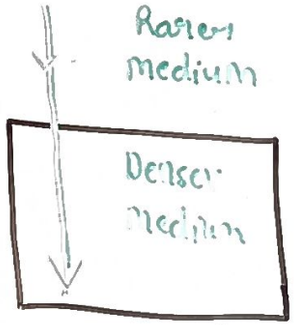
2. Denser medium → (Light.) → Rarer medium.

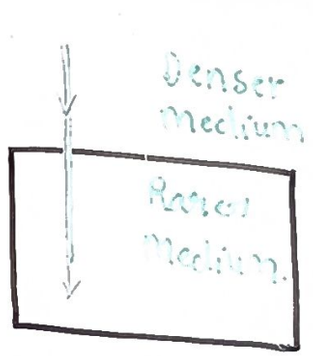


* When light travels from denser medium to rarer medium, it bends away from normal.

* $r > i$

3. Light travels normally / perpendicularly.



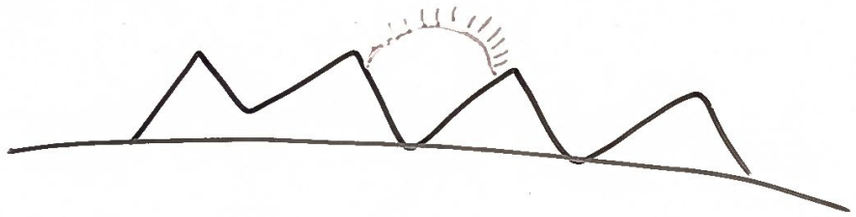


* When ligh travels along the normal from one transparent medium to another transparent medium it does not bend & passes straight through with no bending.

NATURAL PHENOMENA DUE TO REFRACTION.

1. Early sunrise & early sunset.

① **Sunrise:-** The appearance of sun above the horizon.



② As we go deeper inside the atmosphere, the density of the air increases.

③ Thereby, the RI of atmospheric air also increases as we go deeper/down the atmosphere.

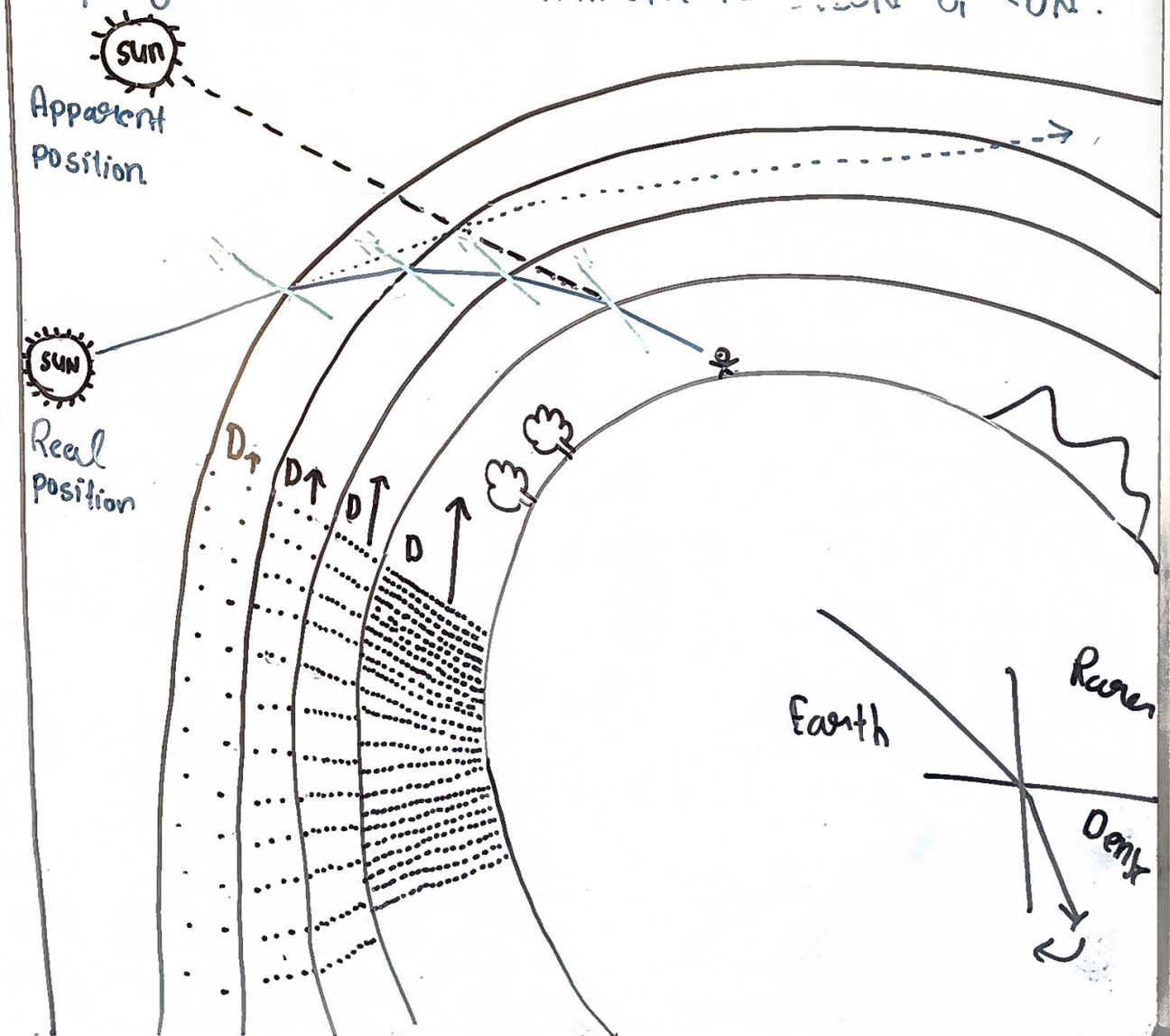
④ The light emitting from the sun travels in straight path in space.

⑤ As soon as light enters the earth's atmosphere, it gradually passes from low density air (low RI) to high density air (high R.I.).

⑥ Thereby, light rays continuosly bend toward the normal as it travels through atmosphere.

⑦ Thus, the sun appears slightly above it's real position & we see early sunrise & early sunset.

Apparent position of sun :- The pseudo position of the sun above its real position due to refraction of light is called 'APPARENT POSITION OF SUN'.



MIRAGE

① Defn:- It is an optical illusion where the image of an object formed inside the ground / road / desert.

② The air near the hot surface (Road, desert) becomes hot & cool down as we go above.

③ The density of hot air is relatively lower than the cold air. (less hot air).

④ Thereby, the R.I. of hot air near the hot surface is lower & R.I. keep increasing as we go above the hot surface.

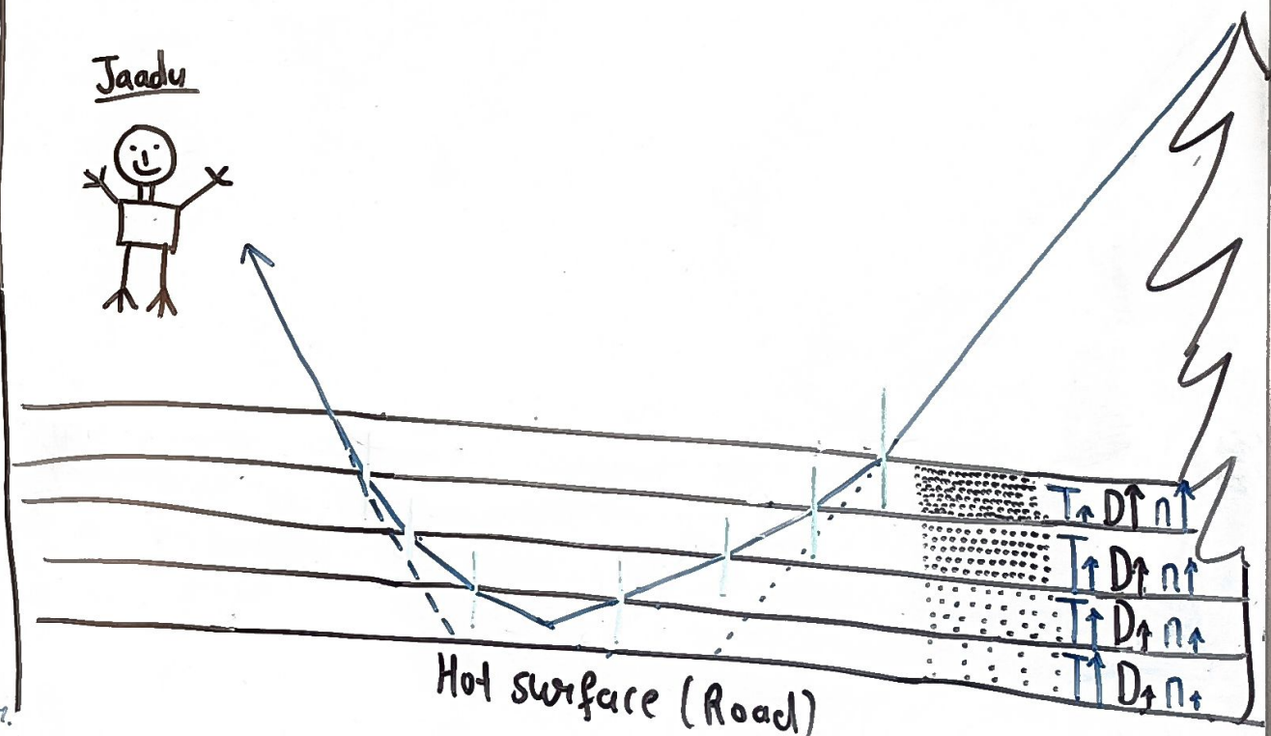
⑤ The light rays coming from the distant object travels from denser air (more R.I.) to rarer air (low R.I.)

⑥ Thus light rays bends away from normal as it passes from denser medium to rarer medium.

⑦ Near the hot surfaces, the light shows TOTAL INTERNAL REFRACTION & light bounce back.

⑧ The we perceive the light of the distant object coming from the ground.

⑨ But this is not the real object but the image of the object & we called it MIRAGE.



TWINKLING OF THE STARS.

① Although the stars are very large celestial body, they appear as point source in the night sky because they are at very-very large distance from the earth.

② As we go deeper inside the earth's atmosph. the density of air increases gradually & so R.I.

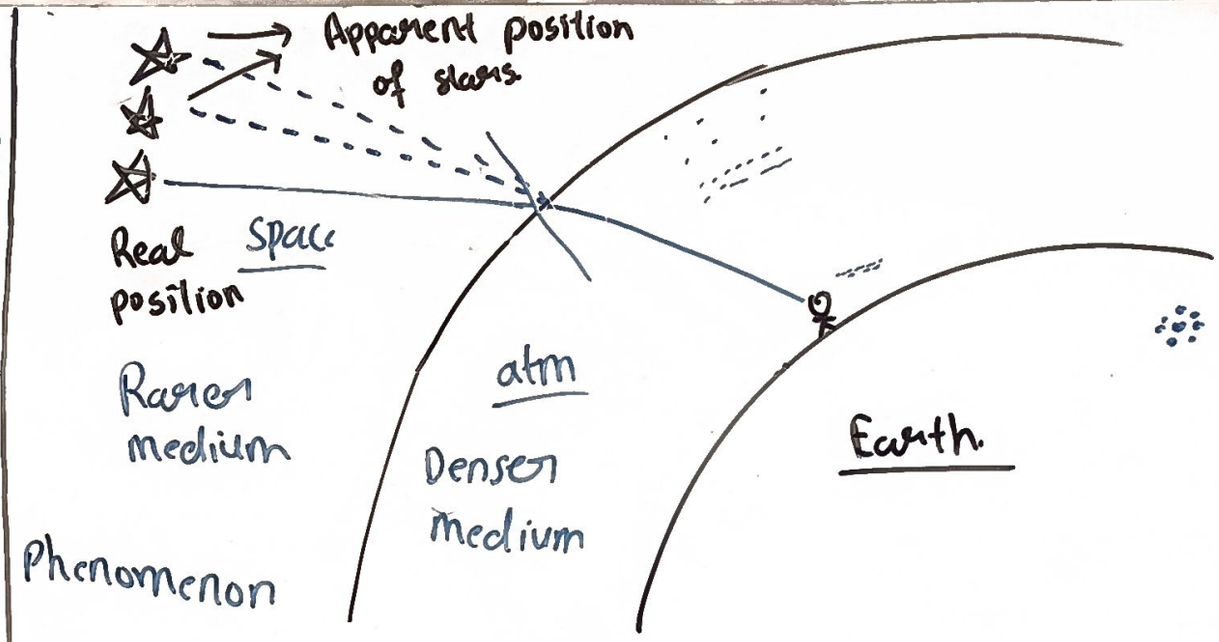
③ When starlight enters the earth's atmosphere, it bends toward the normal as it travels from rarer medium to denser medium.

④ Along with this, the atmospheric air is not static it flows continuosly.

⑤ Thereby, the temperature & density of the air also fluctuate/change & so R.I.

⑥ Due to all these phenomena occuring in atmospheric air, the position & brightness of stars change/fluctuate continuosly & randomly.

⑦ This results in 'TWINKLING OF STARS.'



Phenomenon

WHY PLANETS DO NOT TWINKLE.

① Since the planets are relatively close to earth as compare to stars, it appears large i.e. collection of point source.

② Due to the phenomena of
i) Changing R.I. of atmospheric air
ii) Flowing of atmospheric air
iii) Changing temperature & density of atmospheric air.

The position & brightness of each point sources from the planet change continuosly & randomly.

③ However the average position & brightness of the planet remain same over the course of time.

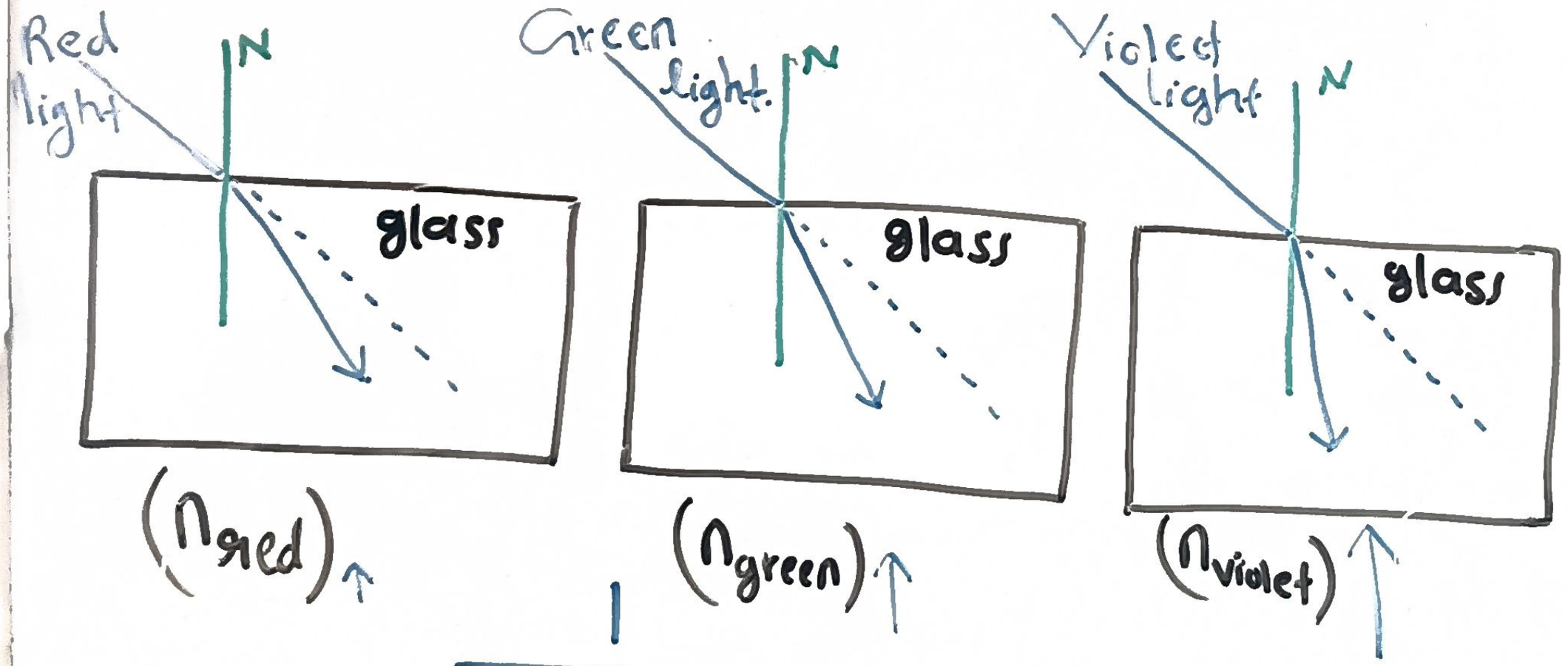
④ Thereby planets DO NOT twinkle.

Refractive Index.

① is different for different media.

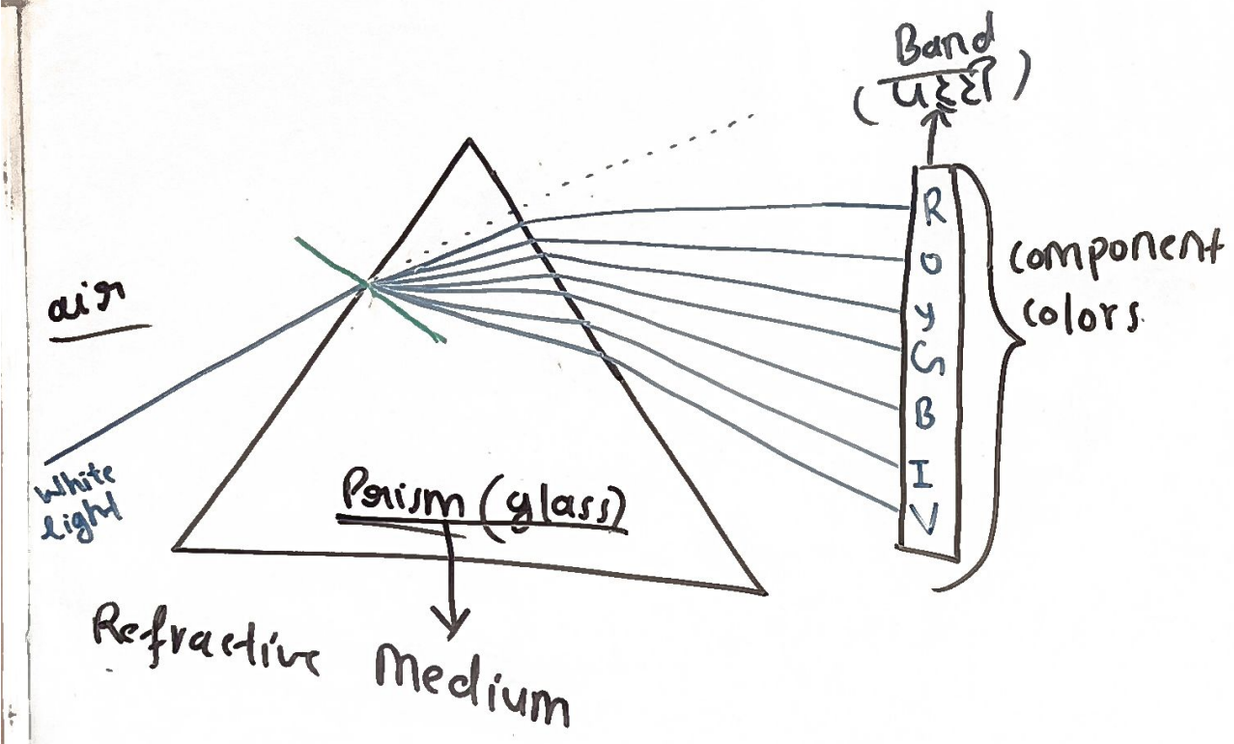
eg.→ $n_{air} = 1.0003$, $n_{H_2O} = 1.31$, $n_{ice} = 1.33$,

② is different for different colours of light in the same medium.



$(n_{red})\uparrow$ $(n_{green})\uparrow$ $(n_{violet})\uparrow$

③ $\uparrow$ R.I. $\downarrow \propto \uparrow$ Speed of light $\downarrow$



DISPERSION OF LIGHT.

*Defn:- The process of separation of white light into it's component colours (VIBGYOR) when it passes through any refractive medium.

*Spectrum:- The band of the distinct colors obtained by separation of white light into its component colors while passing through any refractive medium.

REASON OF DISPERSION OF LIGHT

① The R.I. of different colors [diff. wavelength (λ)] of light is different in the same medium.

② The white light is the mixture of seven different colors of light i.e.

i) Violet → n ↑ → Maximum deviation (सबसे ज्यादा मुड़ेंगी)
ii) Indigo → n ↑
iii) Blue → n ↑
iv) Green → n ↑
v) Yellow → n ↑
vi) Orange → n ↑
vii) Red. → n ↑ → Least deviation (सबसे कम मुड़ेंगी)

③ Each of these componet lights have different R.I.

④ Thus when the white light passes through any refractive medium, then its different component light (VIBGYOR) deviate differently.

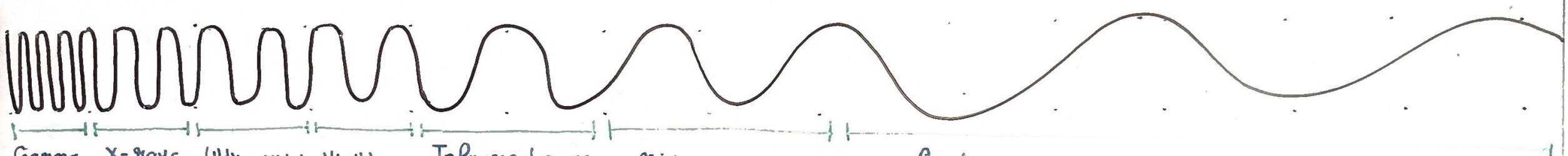
⑤ Thus, each of the component light of white light separate when passes throug the refractive medium.

⑥ This causes "DISPERSION OF LIGHT".

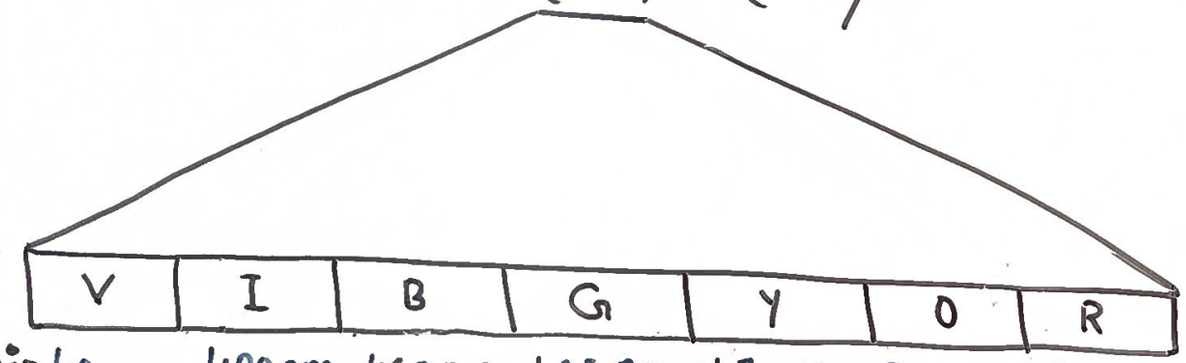
DISPERSION OF LIGHT

① Discovered by Sir Isaac Newton.

Electromagnetic Spectrum.



Gamma ray (≤ 10pm) | X-rays (10pm – 10nm) | Ultra-Violet rays (10nm – 400nm) | Visible light/rays (400nm – 700nm) | Infrared rays (700nm – 1mm) | Micro wave (1mm – 1m) | Radio wave. (1m – thousands of km)

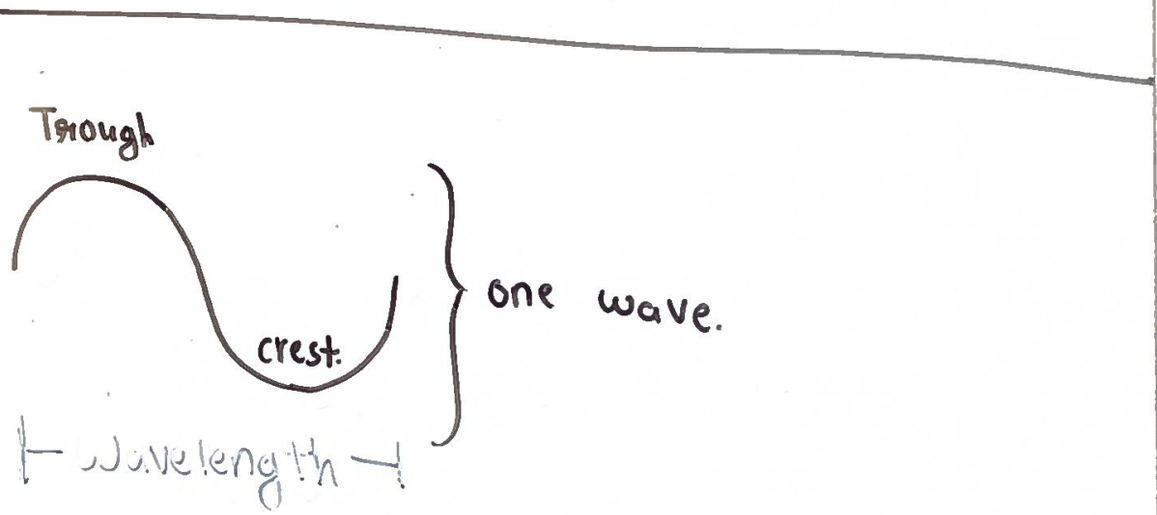


λ :- 400nm (Shortest) | 400nm – 450nm | 450nm – 495nm | 495nm – 570nm | 570nm – 590nm | 590nm – 620nm | 700nm longest.

n :- ↑ ↑ ↑ ↑ ↑ ↑ ↑

Bending :- Maximum (most) ... minium. (lest)

$n \propto \frac{1}{\lambda}$ — conclusion.



Some value to remember

$1\ nm = 10^{-9}\ m$ ⟶ nano meter

$1\ pm = 10^{-12}\ m$ ⟶ pico meter

$c = 3 \times 10^{8}\ m/s$ ⟶ speed of light in vacuum.

* Denser medium —LIGHT.→ Rarer medium

1. PARTIAL REFLECTION.

Defn:- i) Part of light rays reflect back to the denser medium.

ii) Remaining light rays refract (bend) into the rarer medium.

→ This is called 'PARTIAL REFLECTION'.

2. CRITICAL ANGLE.

① $i < r$

② $i(\uparrow) \Rightarrow r(\uparrow)$

③ Defn:- → For a specific value of incidence angle (i), the refracted angle (r) becomes 90°.

→ This value of incidence angle is called CRITICAL ANGLE.

Que All incidence angles are critical angle; but all critical angles are not incrdence angle.

→ True

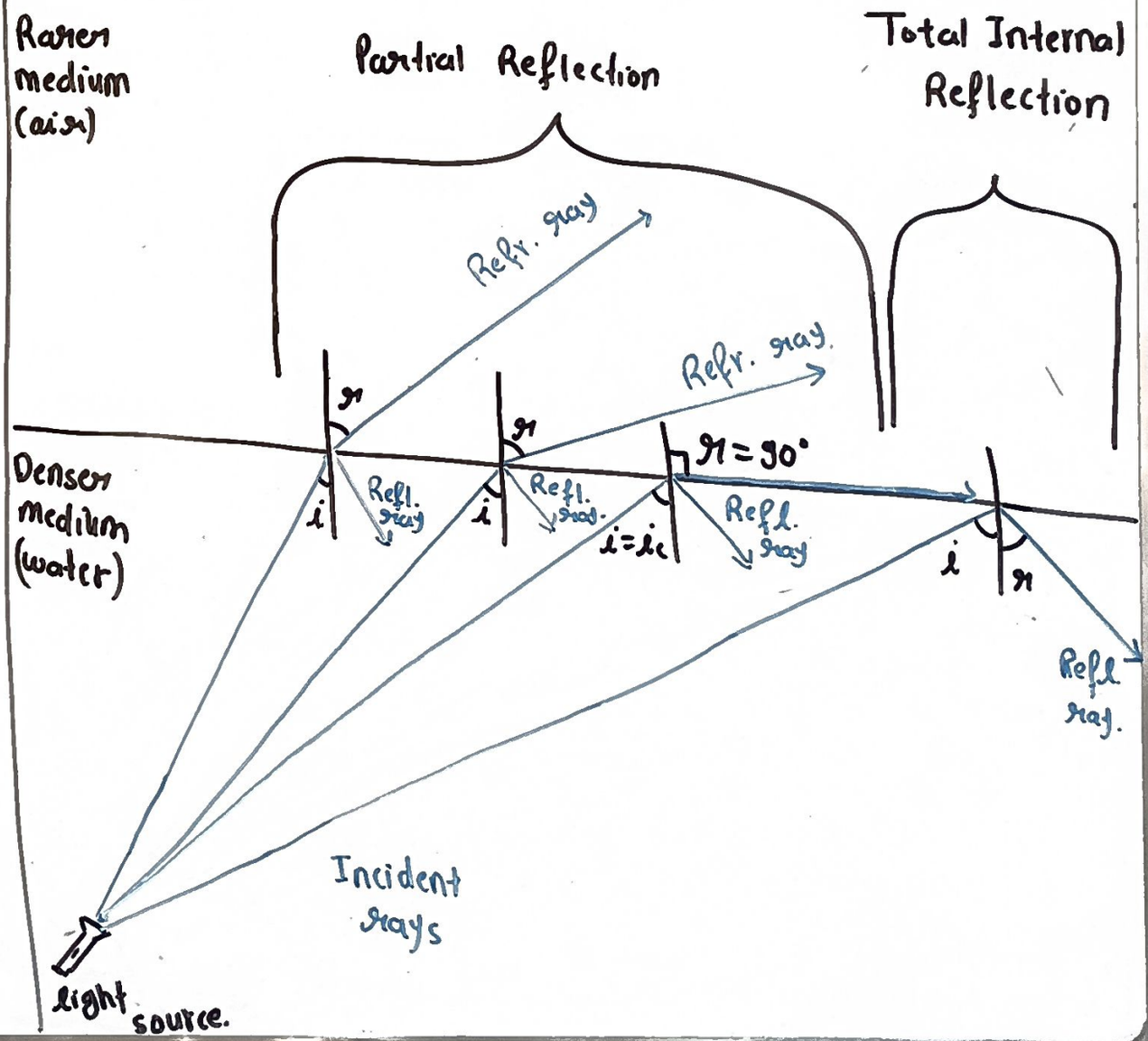
→ false

3. TOTAL INTERNAL REFLECTION.

Defn:- i) Condition :- $i > i_c$

ii) All the light reflected back into the denser medium.

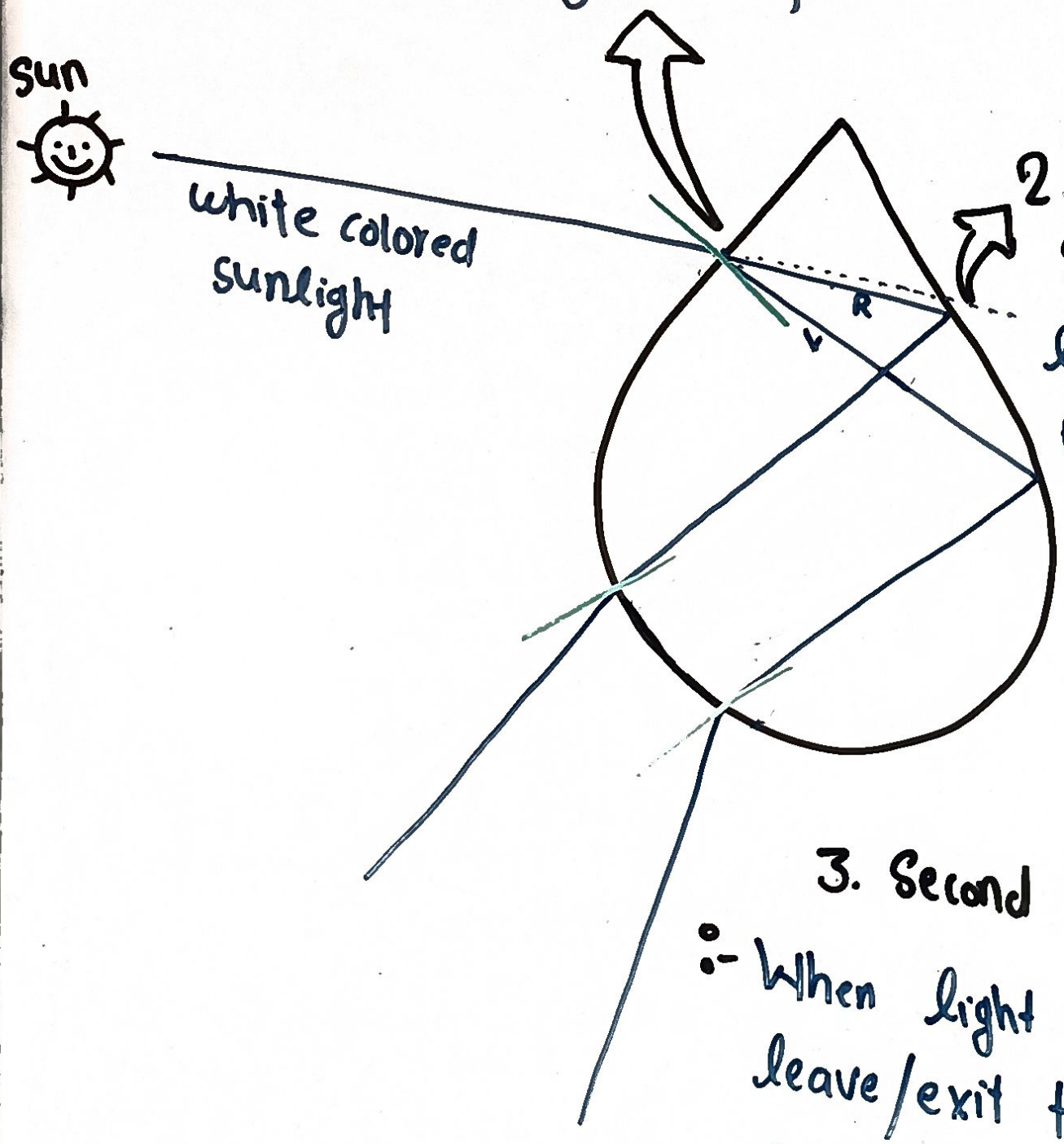
→ This is called TOTAL INTERNA REFLECTION.



FORMATION OF RAINBOW.

1. Refraction + Dispersion

:- When sunlight enters the water droplets, it gets refracted as well as dispersed.



2. Total Internal Refl.

:- Inside the droplet, light rays (All 7 colors) undergo TIR.

3. Second Refraction.

:- When light rays (All 7 colors) leave/exit the water droplet, it undergoes second refraction.

Conclusion:- As the result of all these 3 phenomena occuring across the water droplet, we see the beautiful natural phenomena knows as Rainbow